
EDUCATION

UNIVERSITY OF VIRGINIA

Charlottesville, VA

May 2027

- B.S. in Data Science, B.A. in Computer Science
- GPA: 3.95/4.0
- Relevant courses: Data Structures and Algorithms II, Multivariable Calculus, Linear Algebra, Advanced Databases, Computational Probability, Machine Learning II

RESEARCH & WORK EXPERIENCE

NASA JET PROPULSION LABORATORY (JPL)

Jun. 2026 - Present

Student Researcher

Pasadena, CA

- Fine-tuning time-series foundation models for near real-time forecasting-based anomaly detection on telemetry data using ~100 channels from Sentinel-6A.
 - Leveraging parameter-efficient fine-tuning (PEFT) to adapt a pretrained model MOMENT to mission-specific operational data across channels. Evaluated against two baseline models: cumulative sum control chart (CUSUM) and a 1-D convolutional neural network (1D-CNN). Measured anomaly detection accuracy with custom-made metrics such as adjusted F1-score, R-based-F1, and latency. Demonstrated high detection accuracy and low false-negative rates for real adoption and integration into current operations tools.
 - Tested a side-by-side comparison of data imputation techniques: forward-filling, Kalman filter, seasonal lookahead, MOMENT-reconstruction and MOMENT-forecasting to enhance data preprocessing pipelines. **Improved model accuracy by 19% after filling in data gaps.**

SCOTT DONEY COMPUTATIONAL BIOGEOCHEMISTRY RESEARCH LAB

Aug. 2023 - Present

Undergraduate Student Researcher

Charlottesville, VA

- Implemented 4 ML models: multiple linear regression, random forest regression, XGBoost, and LSTM models for complex multivariate time series forecasting.
 - Built end-to-end ML pipelines in UVA cloud computing cluster using TensorFlow, including feature engineering, hyperparameter optimization via GridSearch, and rigorous model validation using R^2 , RMSE, and MAE metrics.
 - Wrangling **large-scale time-series datasets of 30+ years** to generate robust forecasts of primary productivity. Research manuscript currently undergoing peer review.

MONTEREY BAY AQUARIUM RESEARCH INSTITUTE (MBARI)

Jun. 2025 - Aug. 2025

Student Researcher

Monterey Bay, CA

- Selected as 1 of 14 students out of 800+ applicants. Developed a novel benchmark to assess YOLOv8 object detection and ByteTrack multi-object tracking on deep-sea video footage.
 - Constructed a gold standard human-annotated dataset containing 4 videos and compiled 10 evaluation metrics, i.e. HOTA, to quantify model performance. **Achieved 85% detection and tracking accuracy.**
 - Built a documented analytics workflow and codebase in Python to assess dataset quality and detect/resolve issues in training data, which **accelerated video annotation/labeling process 10x.**

THE MITRE CORPORATION

Jun. 2022 - Aug. 2024

Student Researcher

McLean, VA

- Implemented forecasting models to investigate AI-driven approaches for climate resilience and informed decision-making. Built automated ETL pipelines to ingest, process, and organize data and leveraged relational databases, SQL query optimization, and efficient resource management to support quick metadata retrieval. Designed and implemented a data visualization platform for stakeholders using AWS Lambda, EC2, and S3.

PROJECTS & EXTRACURRICULARS

OCEANX SCIENCE IMPACT CHALLENGE

Oct. 2025 - May 2026

- Selected as a finalist in the OceanX Science Impact Challenge and developed an end-to-end video analysis platform. Integrated **YOLOv11 object detection and BioCLIP-2 image embeddings** for zero-shot species classification. Ran multi-container applications using Docker Compose for reproducible deployments and streamlined execution.

SOLve

Jun. 2019 - Aug. 2023

- Supervised a team of 50 people in developing a mobile application, SOLve, to help students prepare for the Virginia SOLs, standardized tests that measure learning for students in grades 3-11.

SKILLS & CERTIFICATIONS

Publications: DeepSea MOT: A Benchmark Dataset for Multi-Object Tracking on Deep-Sea Video ([arXiv.2509.03499](https://arxiv.org/abs/2509.03499))**Programming Languages:** Python, Java, SQL, C, MATLAB, R**Relevant Skills/Frameworks:** Github, Tableau, Power BI, Docker, MongoDB**Certifications:** AWS Cloud Practitioner (2025), NAUI Open Water Scuba Diving (2025)